

Manufacturing & Video-Analytics Intelligent Report

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Executive Summary

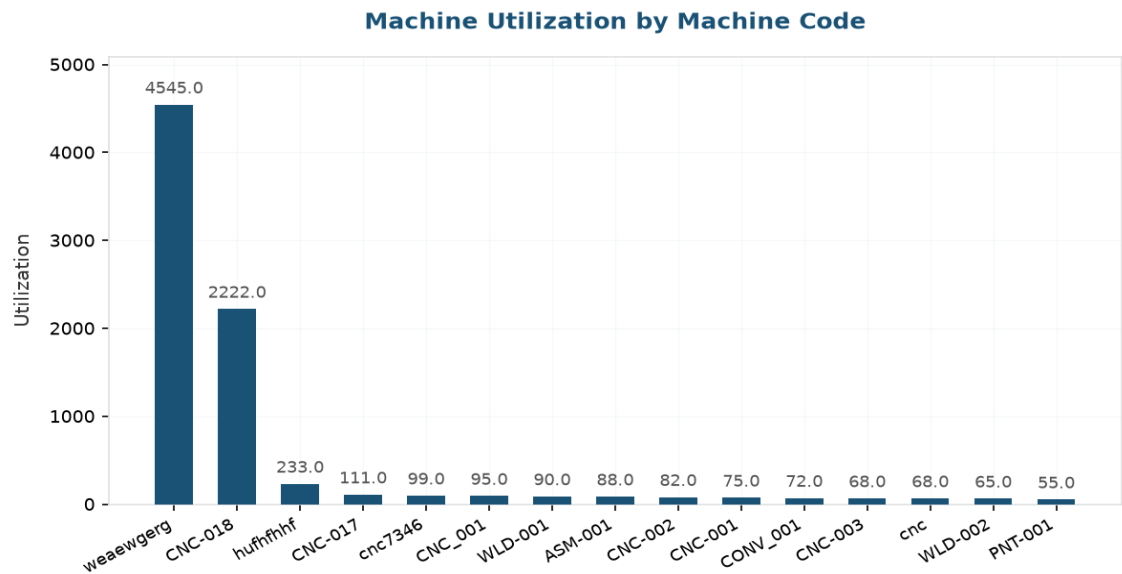


Matched keywords: machine, utilization

Telemetry window: 2026-09-24T09:35:37Z → 2026-09-25T09:35:37Z (bucketed every 5m)

Machine Fleet & Master Utilization Analysis

The MachineMaster dataset contains 28 machines across various categories such as CNC Lathes, CNC Mills, and Welders. The recorded utilization metrics show an average of 296.93%, which is heavily skewed by an extreme outlier of 4545.0% on at least one machine, indicating a potential data logging or scaling error. Similarly, machine capacity averages 909.07 units but contains an outlier of 10454.0 units. Excluding these anomalies, the fleet represents a diverse set of manufacturing assets. Two machines report 0% utilization, representing idle assets that require operational review.

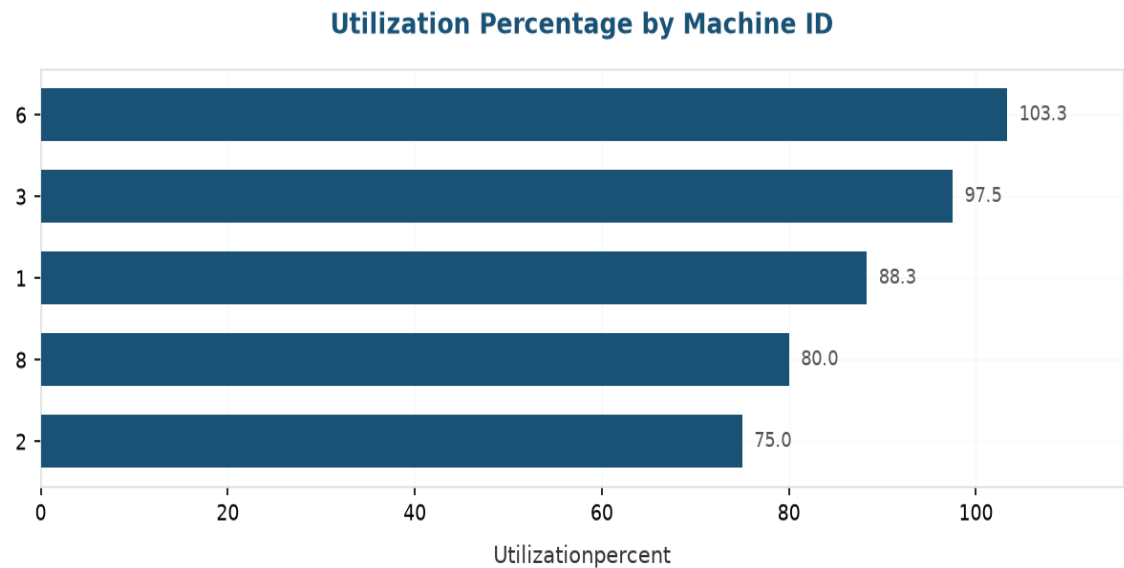


Data preview — MachineMaster (showing 10 of 28 points/rows)

MachineCode	Machineld	MachineName	Utilization	MachineType	Capacity	CapacityPerHour	CustomerId
CNC-001	1	CNC Lathe #1	75.00	CNC Lathe	100.00	10.00	13
CNC-002	2	CNC Lathe #2	82.00	CNC Lathe	100.00	10.00	13
CNC-003	3	CNC Mill #1	68.00	CNC Mill	80.00	8.00	13
WLD-001	4	MIG Welder #1	90.00	MIG Welder	50.00	6.00	13
WLD-002	5	TIG Welder #1	65.00	TIG Welder	40.00	4.00	13
ASM-001	6	Assembly Station #1	88.00	Manual Station	60.00	5.00	13
CONV_001	7	Assembly Station #2	72.00	Manual Station	60.00	5.00	13
CNC_001	8	Auto Assembly Line	95.00	Automated	200.00	20.00	13
PNT-001	9	Paint Booth #1	55.00	Paint Booth	30.00	4.00	13
rere	10	ert	30.00	rreg	0.00	15.00	13

Capacity Analysis & Shift-Level Utilization

The CapacityAnalysis dataset provides a more realistic operational view across 11 records, showing machine utilization percentages ranging from 60.0% to 120.0%, with a healthy overall average of 90.91%. This indicates high asset utilization, with some machines operating above 100% capacity (up to 120.0%), suggesting potential bottlenecks or overtime work. The data spans multiple shifts (Shift IDs 1, 2, and 4) for 5 unique machines. Managing these high-utilization assets is critical to prevent equipment wear and scheduling conflicts.



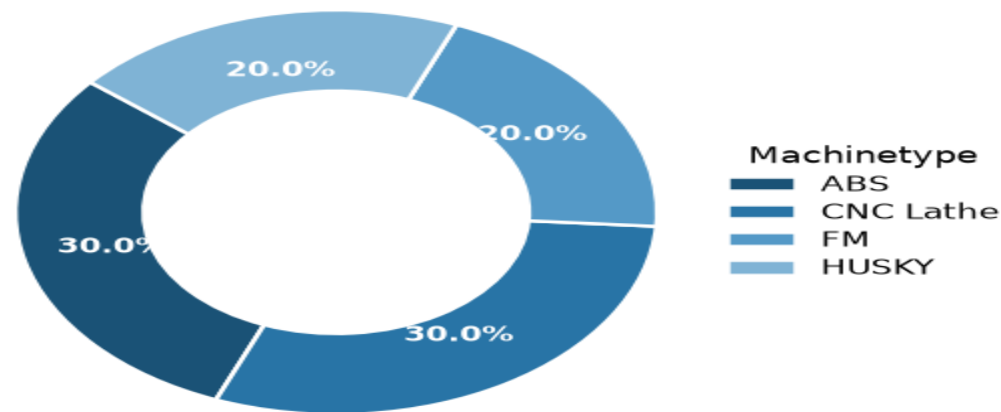
Data preview — CapacityAnalysis (showing 10 of 11 points/rows)

Machineld	UtilizationPercent	CapacityId	CustomerId	ShiftId
1	115.00	16	13	1
2	80.00	17	13	1
3	95.00	18	13	2
6	110.00	19	13	1
8	80.00	20	13	4
1	90.00	21	13	1
2	70.00	22	13	1
3	100.00	23	13	2
6	80.00	24	13	1
1	60.00	25	13	1

Machine Fleet Specifications and Material Mapping

The core Machine table lists 10 primary assets categorized into types like ABS, FM, HUSKY, and CNC Lathe, with capacities ranging from 120 to 250 (averaging 148.57). These machines are mapped to finished goods (FG) and raw materials (RM) through 39 FG mapping records and 8 material mapping records. For instance, machines are linked to specific preforms and packaging types (e.g., Pet Preform Clear Bags and Boxes). This mapping ensures traceability and proper routing of materials to the correct machine types during production runs.

Distribution of Machine Types



Data preview — Machine (showing 10 of 10 points/rows)

MachineCode	Machineld	MachineName	MachineType	Capacity	CompanyId	FGName	QRCodeUrl
ABS-01	1	ABS-01	ABS	120	None	None	https://localhost:7204//Do C...
FM-01	2	FM-01	FM	120	None	None	https://localhost:7204//Do C...
HUSKY-02	3	HUSKY-02	HUSKY	120	None	None	https://localhost:7204//Do C...
ABS-02	4	ABS-02	ABS	150	None	None	https://localhost:7204//Do C...
ABS-03	5	ABS-03	ABS	160	None	None	https://localhost:7204//Do C...
HUSKY-01	6	HUSKY-01	HUSKY	250	None	None	https://localhost:7204//Do C...

MachineCode	Machineld	MachineName	MachineType	Capacity	CompanyId	FGName	QRCodeUrl
FM-02	7	FM-02	FM	120	None	None	https://localhost:7204//Doc...
CNC_001	8	CNC_001	CNC Lathe	Thermopac FX 600	13	415	260.00
CONV_001	9	CONV_001	CNC Lathe	LT-20C	13	415	80.00
CONV_002	10	CONV_002	CNC Lathe	LT-20C	13	415	80.00

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
MachineMaster	28	95.1%	2	Fair
CapacityAnalysis	11	100.0%	0	Good
Machine	10	82.5%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
WeighingMachine	7	100.0%	0	Good

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,088	3,475	8,563	\$0.0389